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Final project:
Yacht Hydrodynamic Resistance

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1 Data and Problem

We analyse the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1, 2]: 22 sailing-yacht hull forms tested at several speeds, for $n = 308$ observations. Each row holds five hull *form factors* — LCB, **prismatic**, L_disp, beam_draught and L_beam (constant within a hull) — and the dimensionless **Froude number** $Fn = v/\sqrt{gL} \in [0.125, 0.45]$; the response is the residuary resistance per unit weight of displacement (Table 1).

For slender hulls, physics predicts a power law resistance $\propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. We study it in a Bayesian framework:

1. log–log regression to estimate and *verify* $\hat{\beta}_F \approx 4$;
2. BAS/BMA over the 2^5 hull-factor submodels, to see which carry signal once Fn is included;
3. a heteroscedasticity-corrected nonlinear model on the retained factors, contrasted with the log–log fit;
4. posterior predictive curves, with and without the heteroscedasticity treatment, against the data.

2 Log–log Regression

2.1 Model and prior

The physics of slender sailing hulls suggests that, in the bulk regime, the residuary resistance grows as a power of the **Froude number**, resistance $\propto Fn^{\gamma_F}$ with γ_F close to 4. Taking logarithms turns this multiplicative law into a linear model, in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors. Since the smallest observed resistance is $0.01 > 0$, the logarithm is well defined for every record and no offset is required. Writing the $n = 308$ observations in matrix form as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\beta} \in \mathbb{R}^p$, $p = 7$, we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \propto 1, \quad \pi(\sigma^2) \propto \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference and provides a transparent Bayesian baseline against which the model-selection and heteroscedasticity-corrected analyses of the following sections can be compared.

2.2 Posterior and credible intervals

Under the prior (2) the posterior is Normal–Inverse–Gamma and is available in closed form. With $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ the least-squares estimate and $s^2 = \|\mathbf{y} -$

$\mathbf{X}\hat{\boldsymbol{\beta}}\|^2/(n-p)$ the unbiased error variance,

$$\sigma^2 \mid \mathbf{y} \sim \text{Inv-Gamma}\left(\frac{n-p}{2}, \frac{(n-p)s^2}{2}\right), \quad (3)$$

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}\left(\hat{\boldsymbol{\beta}}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right). \quad (4)$$

Marginalising (4) over σ^2 yields a multivariate Student- t posterior for $\boldsymbol{\beta}$, centred at the least-squares estimate $\hat{\boldsymbol{\beta}}$ with scale matrix $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ and $n-p$ degrees of freedom. A direct consequence is that, under this reference prior, the posterior 95% credible intervals are *numerically identical* to the classical OLS confidence intervals, and any posterior probability (such as $\Pr(\beta_F > 4 \mid \mathbf{y})$) follows from the same Student- t . No simulation is therefore required: we read all point estimates, 95% credible intervals and the credible band of the fitted curve directly off the fitted linear model (`lm` and `confint` in R).

2.3 Results

Table 2 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale). The posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \quad 95\% \text{ CI} = [4.60, 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$. Fig. 1 (left) shows the log-log scatter with the posterior-mean line and its 95% credible band: the band is extremely tight, reflecting the 308 observations and the strong signal of the Froude number. The posterior of β_F is displayed in Fig. 1 (right); the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

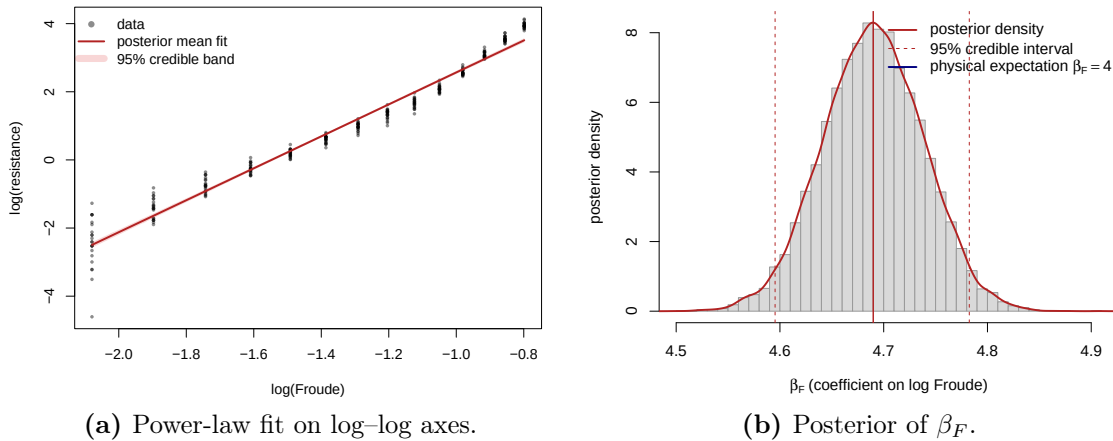


Figure 1: Left: observed $\log(\text{resistance})$ against $\log(\text{Froude})$ with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Student- t posterior density of the Froude exponent β_F ; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 2 and Fig. 3). The **prismatic** coefficient is especially poorly identified, with a posterior standard deviation of about 1.6, because the form factors vary only across the 22 hulls and are strongly collinear. This near-redundancy is precisely what the Bayesian model selection of the next section (BAS/BMA over the 2^5 submodels) is meant to resolve, by quantifying which covariates carry signal through their posterior inclusion probabilities rather than through single-model p -values.

Residual diagnostics. Although the global fit is excellent, the residuals reveal two systematic defects that the simple log–log model cannot capture (Fig. 2). First, the residuals-versus-fitted panel shows a clear U-shaped pattern: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds, consistent with the local Froude exponent increasing from ≈ 4.7 in the full range to above 6 in the high-speed subset. Second, the spread of the residuals is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 1. The Normal Q–Q plot confirms heavier-than-Gaussian lower tails. These two findings (speed-dependent curvature and non-constant variance) directly motivate the heteroscedasticity-corrected nonlinear model developed in the subsequent section.

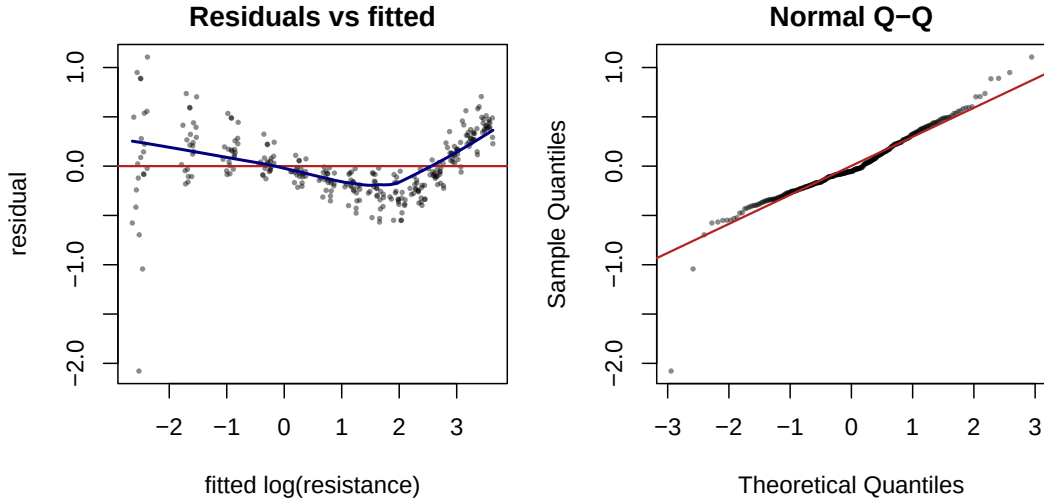


Figure 2: Residual diagnostics for the log–log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q–Q plot, showing heavier-than-Gaussian lower tails.

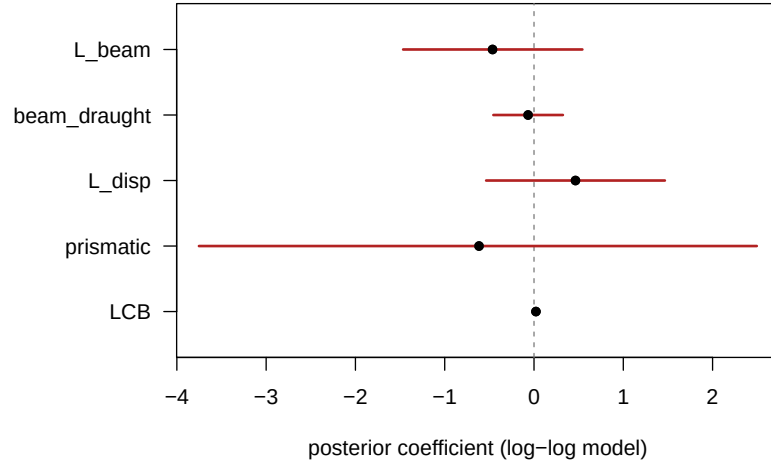


Figure 3: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log-log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

3 BAS

4 Heteroscedasticity-Corrected Nonlinear Model

References

- [1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.
- [2] J. Gerritsma, R. Onnink, and A. Versluis. *Geometry, Resistance and Stability of the Delft Systematic Yacht Hull Series*. Technical Report. Delft University of Technology, Ship Hydromechanics Laboratory, 1981.

A Tables

Table 1: Variables of the yacht hydrodynamics dataset.

Variable	Description
LCB	Longitudinal position of the centre of buoyancy
prismatic	Prismatic coefficient
L_disp	Length–displacement ratio
beam_draught	Beam–draught ratio
L_beam	Length–beam ratio
Froude	Froude number, $Fn = v/\sqrt{gL}$
resistance	Residuary resistance per unit weight of displacement

Table 2: Posterior summary of the log–log model (1) under the reference prior. Point estimates and 95% credible intervals coincide with the OLS fit (`lm/confint`); $\text{Pr}(> 0)$ is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	$\text{Pr}(> 0)$
Intercept β_0	7.183	0.977	5.245	9.122	1.00
$\log(\text{Froude})$ β_F	4.690	0.048	4.595	4.782	1.00
LCB	0.021	0.012	−0.003	0.046	0.96
prismatic	−0.616	1.597	−3.753	2.495	0.35
L_disp	0.464	0.512	−0.536	1.466	0.82
beam_draught	−0.067	0.200	−0.455	0.324	0.37
L_beam	−0.464	0.515	−1.466	0.542	0.18